Mechanism 1
Genetic Algorithm (GA)

structural & behavioral
node design ($Task_1$)

distributions of reliability and efficiency parameters of nodes

GA → DQN

statistical distributions of E, T and reliability parameters initialize the random-event generators of the digital twin

Mechanism 3 — DQN agent

seven operational tasks: $Task_2$–$Task_5$, $Task_7$–$Task_9$

Digital twin (training environment)

stochastic simulation: node topology, material and information flows, reliability distributions

Q-network + target network

Replay buffer experience (s, a, r, s′)

Integral reward $R = \Sigma\, w_i \cdot r_i$ aggregates the partial effects of all seven tasks

DQN → GNN

GNN → DQN

Mechanism 4
Graph Neural Network

topology-aware scaling
($Task_6$)

graph model of the line, load-redistribution recommendations

node loads and topology from solved $Task_2$/$Task_7$ instances form the GNN training set

scaling recommendations return to the action space of $Task_2$ and the state input of $Task_7$

AIC → DQN

real-time state vector s_t synchronizes the digital twin

DQN → AIC

control commands a_t executed via slave modules

Mechanism 2 — Artificial Intelligence Complex (AIC)

edge devices: CNN (spectral features), RNN/LSTM (time series); indirect reliability assessment ($Task_{10}$); master–slave modules

sensor signals

cyber-physical control loop

actuation

Physical production line

nodes, warehouse, material flows

Closed optimization cycle

GA → digital twin → DQN ↔ AIC → GNN with GNN recommendations returned to the DQN action space